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# On the prediction of spray angle of liquid-liquid pintle injectors

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ABSTRACT

The pintle injector is famous for its capability of deep throttling and low cost. However, the pintle injector has been seldom investigated. To get a good prediction of the spray angle of liquid-liquid pintle injectors, theoretical analysis, numerical simulations and experiments were conducted. Under the hypothesis of incompressible and inviscid flow, a spray angle formula was deduced from the continuity and momentum equations based on a control volume analysis. The formula was then validated by numerical and experimental data. The results indicates that both geometric and injection parameters affect the total momentum ratio (*TMR*) and then influence the spray angle formed by liquid-liquid pintle injectors. *TMR* is the pivotal non-dimensional number that dominates the spray angle. Compared with gas-gas pintle injectors, spray angle formed by liquid-liquid injectors is larger, which benefits from the local high pressure zone near the pintle wall caused by the impingement of radial and axial sheets.

## 1. Introduction

The pintle injector is a unique propellant injection device compared with traditional injectors [1]. It was first applied in rocket engines about 50 years ago. To date, it has been applied into rocket engines over a wide thrust range and with 25 different propellant combinations [2]. One famous pintle injector engine is the Lunar Module Decent Engine (LMDE), which delivered astronauts safely to the surface of the moon for the first time. Until recently, Northrop Grumman was pursuing liquid oxygen/hydrogen pintle injector technology for the descent stage of the lunar exploration [3]. Though the pintle injector was initially designed for variable thrust engine because of the deep throttling capacity [4,5], it was also applied into fixed thrust engine later due to its excellent performance, such as TR-106 [6] and Merlin series [7]. The Merlin engine became famous recently, because it helped SpaceX recycle its Falcon 9 rocket on the sea.

The pintle injector possesses of various advantages [2]. First advantage is the deep throttling capacity [5]. A pintle injector engine can adjust its thrust in a wide range, which is necessary for a rocket-powered spacecraft to perform hovering and soft landing. For example, LMDE is qualified for a 10:1 throttle range of 46.7 kN–4.67 kN in a vacuum. Second advantage is the intrinsic combustion stability. The pintle injector engine has never encountered any combustion instabilities in history. Absence of combustion instabilities evidently reduces the development cycle time and costs. Third advantage is its simpleness but high reliability. Generally, a thrust chamber occupies a single pintle injector. The simpleness makes it ease in manufacturing at low cost.

Despite of its advantages and wide applications, studies focusing on a pintle injector are rather limited. Formerly TRW Inc. conducted a large mount of work on pintle injector engines [2], but most work were undisclosed. In 1990s, Mueller and Dressler [8] tested a 178 kN pintle injector engine and gained lots of experience on design. Based on those experiences, they built a 2893 kN pintle injector engine and achieved both high performance and low cost [6]. Later, Austin et al. [9] tested the performance of a pintle engine with nontoxic hypergolic bipropellants. Bedard et al. [10,11] also conducted experiments to study the influence of the pintle injector to thrust chambers. Sakaki et al. [12] captured the spray and flame structure of a planar pintle injector through optical measurements, and compared the spray structure of the fuel-centered configuration and oxidizer-centered configuration. Experiments with an axisymmetric combustor were also conducted by the same authors [13]. They found that the *TMR* had significant influence on the performance of the combustor. Son et al. [14] performed two-dimensional numerical simulations on a fuel centered pintle injector and found that combustion performance was strongly related with recirculation zones inside the combustion chamber. Fang and Shen [15] conducted three-

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**Nomenclature**

| | |
|---|---|
| **A** | surface vector |
| **B** | Body force vector per unit volume |
| $d_p$ | pintle diameter, mm |
| **Fs** | shear force vector |
| $F_{st}$ | surface tension |
| $L_s$ | skip distance, mm |
| $\dot{m}$ | mass flow rate, kg/s |
| **n** | normal unit vector of surface |
| $p$ | pressure, Pa |
| *TMR* | Total momentum ratio |
| $v$ | velocity magnitude, m/s |
| **v** | velocity vector |
| $w_r$ | width of radial slot, mm |
| $w_a$ | width of axial slot, mm |
| **z** | axial unit vector |
| $\alpha$ | volume fraction of liquid |
| $\theta$ | spray angle, deg |
| $\rho$ | density, kg/m$^3$ |
| $\rho_l$ | liquid density, kg/m$^3$ |
| $\rho_g$ | gas density, kg/m$^3$ |

*Subscripts*

| | |
|---|---|
| *1* | control volume 1 |
| *2* | control volume 2 |
| *a* | ambient |
| *ad* | axial direction |
| *c* | collision surface |
| *f* | free surface |
| *i* | inlet surface |
| *l* | left surface |
| *o* | outlet surface |
| *r* | right surface |
| *rd* | radial direction |

dimensional numerical simulations, they found that the *TMR* was a key design parameter and the recirculation zones had big influence on the combustion efficiency.

Compared with hot tests, cold tests are less concerned but attract more attention recently. Boettcher et al. [16] visualized the flow structure of a gas-gas pintle injector by schlieren. They found that the spreading angle of jets increased along with increase of momentum ratio and proposed a formula for the spreading angle prediction. Fang and Shen [15] conducted cold experiments to study the influence of main structure parameters of a gas-liquid pintle injector on atomization cone angles. Their results indicate that the thickness of liquid sheet, injection angle, and skip distance have evidently influence on the atomization cone angle. Son et al. [17,18] studied the discharge coefficient, spray distribution, and spray angle of a gas-liquid pintle injector through numerical simulations and experiments. A empirical correlation, which correlated the spray angle with the momentum ratio and Weber number, was obtained experimentally. Moreover, a design procedure for the pintle injector was developed based on his experiments [19,20]. Rezende et al. [21] conducted some cold tests using different types of liquid-liquid pintle injectors, but only results about the discharge coefficient was reported. Santoro and Merkle [22] investigated the spray of pintle injectors by using liquid oxygen and RP-1. It was found that the spray angle became larger when the momentum ratio increased and the block factor had little effects on the spray angle. Sakaki et al. [23] found that spray angles of liquid-liquid pintle injector were dominated by the *TMR*, but the correlation between the spray angle and *TMR* was not obtained.

As can be seen, few academic studies about the pintle injector have been reported [1,12]. The spray formed by the pintle injector is rarely investigated, and the related knowledge is rather limited. For example, the spray angle, which is an important parameter for characterizing the spatial distribution of the spray, is not well predicted. Therefore, in the present study, a theoretical prediction model of the spray angle is established for liquid-liquid pintle injectors firstly. Then, the model is validated through numerical simulations and experiments and compared with other models. The effects of propellants status on the correlation between the spray angle and momentum ratio are discussed. This paper is organized as follows: section 2 derives the correlation of the spray angle theoretically; section 3 introduced the numerical and experimental methods; section 4 presented the results and discussion.

## 2. Theoretical analysis

### 2.1. Simplification

In a bipropellant pintle injector engine, one of the propellants flows through the pintle and is ejected radially through a series of holes or slots near the tip of the pintle, as shown in Fig. 1. The other propellant leaves the manifold through an annular slot around the base of the pintle. Then the radial sheet and axial sheet impinge each other, spread out and break up. This process is similar to flow impingements in impinging-jet injectors [24].

The present pintle injector has a continuous slot, therefore the liquid sheets are assumed to be axisymmetric to simplify the problem. Then a slice of liquid is picked out for a control volume analysis, as shown in Fig. 2. This slice of fluid is divided into two control volumes according to different propellants. Main parameters in all control surfaces are labeled in Fig. 2.

### 2.2. Formula derivation

Several basic hypotheses must be declared in order to carry out the theoretical analysis.

1. The flow is steady and incompressible.
2. Velocity vectors at both inlets and outlets are normal to the surface.
3. The inlet of control volume 1 and outlets of two control volumes are assumed to be far away from the collision surface. Therefore, the pressures in those surfaces can be treated as ambient pressure.
4. Body force is neglected, such as gravity and electromagnetism etc.
5. No heat transfer and phase change, such as evaporation [25], are considered.
6. Surface tension is ignored.

Then we begin with the momentum equation. The momentum

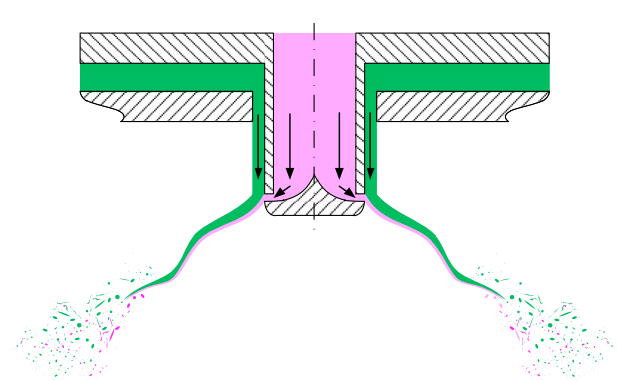

**Fig. 1.** Pintle injector concept.

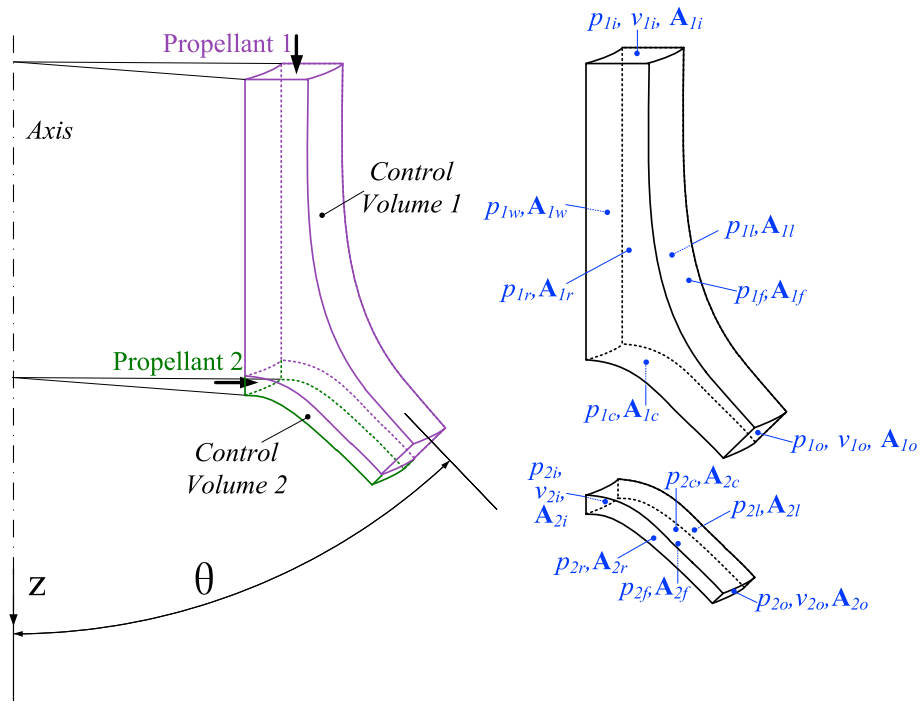


**Fig. 2.** Sketch of control volumes.

equation for a control volume $V$ can be expressed as [26]:

$$\int_V \rho \mathbf{B} dV - \int_A p d\mathbf{A} + \mathbf{F}s = \int_V \frac{\partial(\rho \mathbf{v})}{\partial t} dV + \int_A \mathbf{v}(\rho \mathbf{v} \cdot d\mathbf{A}) \tag{1}$$

Neglecting the body force and unsteady term, Equation (1) reduces to:

$$\int_A \mathbf{v}(\rho \mathbf{v} \cdot d\mathbf{A}) + \int_A p d\mathbf{A} = \mathbf{F}s \tag{2}$$

When the velocity is locally parallel to the wall and phase interface, we obtain $\mathbf{v} \cdot d\mathbf{A} = 0$. By neglecting the friction in the wall and free surface, the momentum equations for the two control volume can be written as follows:

$$\begin{aligned} &\rho_1 \left( \mathbf{n}_{1i} \cdot \int_{A_{1i}} v_{1i}^2 \cdot dA + \mathbf{n}_{1o} \cdot \int_{A_{1o}} v_{1o}^2 \cdot dA \right) + p_{1i} \cdot \mathbf{A}_{1i} + p_{1w} \cdot \mathbf{A}_{1w} + p_{1f} \cdot \mathbf{A}_{1f} \\ &+ p_{1c} \cdot \mathbf{A}_{1c} + p_{1o} \cdot \mathbf{A}_{1o} + p_{1r} \cdot \mathbf{A}_{1r} \cdot + p_{1l} \cdot \mathbf{A}_{1l} = \mathbf{F}s_{1c} \end{aligned} \tag{3}$$

$$\begin{aligned} &\rho_2 \left( \mathbf{n}_{2i} \cdot \int_{A_{2i}} v_{2i}^2 \cdot dA + \mathbf{n}_{2o} \cdot \int_{A_{2o}} v_{2o}^2 \cdot dA \right) + p_{2i} \cdot \mathbf{A}_{2i} + p_{2c} \cdot \mathbf{A}_{2c} \\ &+ p_{2f} \cdot \mathbf{A}_{2f} + p_{2o} \cdot \mathbf{A}_{2o} + p_{2r} \cdot \mathbf{A}_{2r} + p_{2l} \cdot \mathbf{A}_{2l} = \mathbf{F}s_{2c} \end{aligned} \tag{4}$$

As mentioned previously, pressure at the free surface, outlets for both control volume, and inlet of control volume 1 are all equal to the ambient pressure, which means $p_{1i} = p_{1o} = p_{1f} = p_{2o} = p_{2f} = p_a$. Substituting them to Equations (3) and (4) yields:

$$\begin{aligned} &\rho_1 \left( \mathbf{n}_{1i} \cdot \int_{A_{1i}} v_{1i}^2 \cdot dA + \mathbf{n}_{1o} \cdot \int_{A_{1o}} v_{1o}^2 \cdot dA \right) + p_a \cdot \mathbf{A}_{1i} + p_{1w} \cdot \mathbf{A}_{1w} + p_a \cdot \mathbf{A}_{1f} \\ &+ p_{1c} \cdot \mathbf{A}_{1c} + p_a \cdot \mathbf{A}_{1o} + p_{1r} \cdot \mathbf{A}_{1r} \cdot + p_{1l} \cdot \mathbf{A}_{1l} = \mathbf{F}s_{1c} \end{aligned} \tag{5}$$

$$\begin{aligned} &\rho_2 \left( \mathbf{n}_{2i} \cdot \int_{A_{2i}} v_{2i}^2 \cdot dA + \mathbf{n}_{2o} \cdot \int_{A_{2o}} v_{2o}^2 \cdot dA \right) + p_{2i} \cdot \mathbf{A}_{2i} + p_{2c} \cdot \mathbf{A}_{2c} \\ &+ p_a \cdot \mathbf{A}_{2f} + p_a \cdot \mathbf{A}_{2o} + p_{2r} \cdot \mathbf{A}_{2r} + p_{2l} \cdot \mathbf{A}_{2l} = \mathbf{F}s_{2c} \end{aligned} \tag{6}$$

Then adding Equation (5) to Equation (6) leads to the following equation:

$$\begin{aligned} &\rho_1 \left( \mathbf{n}_{1i} \cdot \int_{A_{1i}} v_{1i}^2 \cdot dA + \mathbf{n}_{1o} \cdot \int_{A_{1o}} v_{1o}^2 \cdot dA \right) + p_a \cdot \mathbf{A}_{1i} + p_{1w} \cdot \mathbf{A}_{1w} + p_a \cdot \mathbf{A}_{1f} \\ &+ p_{1c} \cdot \mathbf{A}_{1c} + p_a \cdot \mathbf{A}_{1o} + p_{1r} \cdot \mathbf{A}_{1r} \cdot + p_{1l} \cdot \mathbf{A}_{1l} \\ &+ \rho_2 \left( \mathbf{n}_{2i} \cdot \int_{A_{2i}} v_{2i}^2 \cdot dA + \mathbf{n}_{2o} \cdot \int_{A_{2o}} v_{2o}^2 \cdot dA \right) + p_{2i} \cdot \mathbf{A}_{2i} + p_{2c} \cdot \mathbf{A}_{2c} \\ &+ p_a \cdot \mathbf{A}_{2f} + p_a \cdot \mathbf{A}_{2o} + p_{2r} \cdot \mathbf{A}_{2r} + p_{2l} \cdot \mathbf{A}_{2l} = \mathbf{F}s_{1c} + \mathbf{F}s_{2c} \end{aligned} \tag{7}$$

Considering that the collision surfaces for the two control volume are exactly overlapped in space, then we have $p_{1c} = p_{2c}$, $\mathbf{A}_{1c} + \mathbf{A}_{2c} = 0$, $\mathbf{F}s_{1c} + \mathbf{F}s_{2c} = 0$. Substituting those to Equation (7) and projecting Equation (7) to the axial direction yields:

$$\begin{bmatrix} \rho_1 \left( \mathbf{n}_{1i} \cdot \int_{A_{1i}} v_{1i}^2 \cdot dA + \mathbf{n}_{1o} \cdot \int_{A_{1o}} v_{1o}^2 \cdot dA \right) + p_a \cdot \mathbf{A}_{1i} + p_{1w} \cdot \mathbf{A}_{1w} \\ + p_a \cdot \mathbf{A}_{1f} + p_a \cdot \mathbf{A}_{1o} + p_{1r} \cdot \mathbf{A}_{1r} + p_{1l} \cdot \mathbf{A}_{1l} \\ + \rho_2 \left( \mathbf{n}_{2i} \cdot \int_{A_{2i}} v_{2i}^2 \cdot dA + \mathbf{n}_{2o} \cdot \int_{A_{2o}} v_{2o}^2 \cdot dA \right) + p_{2i} \cdot \mathbf{A}_{2i} \\ + p_a \cdot \mathbf{A}_{2f} + p_a \cdot \mathbf{A}_{2o} + p_{2r} \cdot \mathbf{A}_{2r} + p_{2l} \cdot \mathbf{A}_{2l} \end{bmatrix} \cdot \mathbf{z} = 0 \tag{8}$$

Simplifying Equation (8) gives the following form:

$$\begin{aligned} &-\rho_1 \cdot \int_{A_{1i}} v_{1i}^2 \cdot dA + \rho_1 \cdot \cos\theta \cdot \int_{A_{1o}} v_{1o}^2 \cdot dA + p_2 \cdot \cos\theta \cdot \int_{A_{2o}} v_{2o}^2 \cdot dA \\ &+ p_a \cdot \left( \mathbf{A}_{1i} + \mathbf{A}_{1f} + \mathbf{A}_{1o} + \mathbf{A}_{2o} + \mathbf{A}_{2f} \right) \cdot \mathbf{z} = 0 \end{aligned} \tag{9}$$

As can be seen in Fig. 2, all surfaces except collision surfaces constitute a closed surface, then $\mathbf{A}_{1i} + \mathbf{A}_{1f} + \mathbf{A}_{1o} + \mathbf{A}_{2o} + \mathbf{A}_{2f} + \mathbf{A}_{2i} + \mathbf{A}_{1w} + \mathbf{A}_{1r} + \mathbf{A}_{1l} + \mathbf{A}_{2r} + \mathbf{A}_{2l} = 0$. Because $\mathbf{A}_{1w}$, $\mathbf{A}_{2i}$, $\mathbf{A}_{1r}$, $\mathbf{A}_{1l}$, $\mathbf{A}_{2r}$, $\mathbf{A}_{2l}$ are perpendicular to axial direction, therefore $(\mathbf{A}_{1w} + \mathbf{A}_{2i} + \mathbf{A}_{1r} + \mathbf{A}_{1l} + \mathbf{A}_{2r} + \mathbf{A}_{2l}) \cdot \mathbf{z} = 0$, so $(\mathbf{A}_{1i} + \mathbf{A}_{1f} + \mathbf{A}_{1o} + \mathbf{A}_{2o} + \mathbf{A}_{2f}) \cdot \mathbf{z} = 0$. Then Equation (9) becomes:

$$-\rho_1 \cdot \int_{A_{1i}} v_{1i}^2 \cdot dA + \rho_1 \cdot \cos\theta \cdot \int_{A_{1o}} v_{1o}^2 \cdot dA + \rho_2 \cdot \cos\theta \cdot \int_{A_{2o}} v_{2o}^2 \cdot dA = 0 \tag{10}$$

Then, we obtain:

$$\cos\theta = \frac{\rho_1 \int_{A_{1i}} V_{1i}^2 \cdot dA}{\rho_1 \int_{A_{1o}} V_{1o}^2 \cdot dA + \rho_2 \int_{A_{2o}} V_{2o}^2 \cdot dA} \tag{11}$$

Actually, the spray angle cannot be predicted from Equation (11) because $v_{1o}$ and $v_{2o}$ are unknown. If we assume that the flow is inviscid, there are no boundary layer and shear layer, and then velocities are uniform in face $\mathbf{A}_{1i}$, $\mathbf{A}_{1o}$ and $\mathbf{A}_{2o}$. Therefore Equation (11) becomes:

$$\cos\theta = \frac{\rho_1 A_{1i} v_{1i}^2}{\rho_1 A_{1o} v_{1o}^2 + \rho_2 A_{2o} v_{2o}^2} \tag{12}$$

As pressure at the free surface is equal to the ambient pressure, then according to Bernoulli's theorem: $p + \frac{1}{2}\rho v^2 = const.$ , we have $v_{1i} = v_{1o}$. Considering the continuity equation:

$$\dot{m}_1 = \rho_1 A_{1i} v_{1i} = \rho_1 A_{1o} v_{1o}, \dot{m}_2 = \rho_2 A_{2o} v_{2o} \tag{13}$$

we have $A_{1i} = A_{1o}$, and substitute (13) into (12) leads to:

$$\cos\theta = \frac{1}{1 + \frac{\dot{m}_2^2}{\dot{m}_1^2} \cdot \frac{\rho_1 A_{1i}}{\rho_2 A_{2o}}} \tag{14}$$

Here, $A_{2o}$ is still unknown, spray angle cannot be predicted. Note that total momentum ratio is defined as:

$$TMR = \frac{\dot{m}_2 v_{2i}}{\dot{m}_1 v_{1i}} = \frac{\dot{m}_2^2}{\dot{m}_1^2} \frac{\rho_1 A_{1i}}{\rho_2 A_{2i}} \tag{15}$$

Then if $A_{2i}$ is approximate to $A_{2o}$, an approximate half spray angle $\theta'$ satisfies:

$$\cos\theta' = \frac{1}{1 + TMR} \tag{16}$$

Collision of liquid sheets near the inlet of control volume 2 makes the pressure distribution at face $\mathbf{A}_{2i}$ non-uniform. According to Bernoulli's theorem, the velocity is also non-uniform. Fig. 3 shows velocity profiles at inlets and outlets. And at free surface $\mathbf{A}_{2f}$, velocities have equal values. This indicates that $v_{2i} \le v_{2o}$ everywhere at face $\mathbf{A}_{2i}$. Then considering continuity Equation (13), we obtain $A_{2i} > A_{2o}$. According to Equation (14) we know $\cos\theta < \cos\theta'$, then $\theta > \theta'$. That means the spray angle predicted by $TMR$ is smaller than spray angle in Equation (14). However, this is deduced under the inviscid assumption. The shear layers between phase interfaces will make velocity non-uniform in $\mathbf{A}_{2o}$, and then may lead to a larger $A_{2o}$. So, $A_{2o}$ may close to $A_{2i}$, and Equation (16) is reasonable.

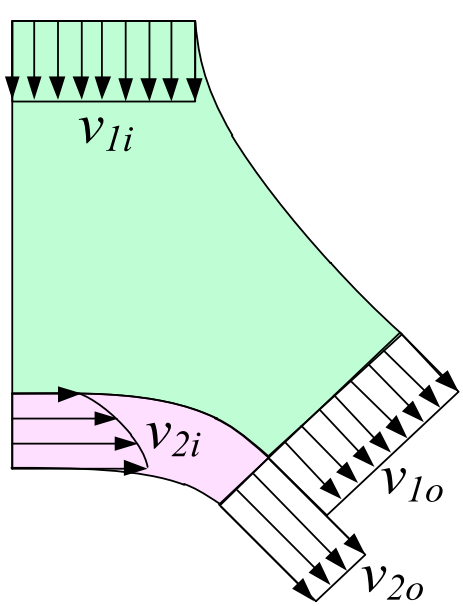


**Fig. 3.** Velocity profiles at inlets and outlets under the assumption of inviscid flow.

## 3. Numerical and experimental methods

### 3.1. Numerical simulations

Fig. 4 shows the configuration of present pintle injector studied in numerical simulations and experiments. Detail dimensional parameters are tabulated in Table 1. The effects of slot dimensions and simulant flow rates are taken into consideration.

In the present study, the problem was to simulate two immiscible liquid sheets impinge on each other near the pintle tip in quiescent air, as shown in Fig. 1. Other sub-processes of combustion [27], such as evaporation, mixing and reaction, were not considered in order to compare the simulation results with cold experiments. Besides, the inner flow inside the pintle was ignored for simplicity.

An Euler-Euler approach (the volume of fluid (VOF) model) is used for the numerical calculation of multiphase flows, which mathematically treats different phases as interpenetrating continua. Ignoring gravity, the incompressible, variable-density, unsteady Navier–Stokes equations with surface tension in the Cartesian form can be represented [28,29] as:

$$\begin{cases} \dfrac{\partial \rho}{\partial t} + \dfrac{\partial \rho u_i}{\partial x_j} = 0 \\ \dfrac{\partial \rho u_i}{\partial t} + \dfrac{\partial \rho u_i u_j}{\partial x_j} = -\dfrac{\partial \rho}{\partial x_j} + \dfrac{\partial}{\partial x_j}\left(\mu \dfrac{\partial u_i}{\partial x_j}\right) + F_{st} \end{cases} \tag{17}$$

For two phases (gas and liquid) flow, the volume fraction of liquid $\alpha$ is used to define the fluid density $\rho$ and dynamic $\mu$ as follows:

$$\begin{cases} \rho = \alpha\rho_l + (1-\alpha)\rho_g \\ \mu = \alpha\mu_l + (1-\alpha)\mu_g \end{cases} \tag{18}$$

The surface tension $F_{st}$ is calculated by using the continuum surface force (CFS) model. $F_{st}$ can be written as the following equation [30]:

$$F_{st} = \sigma \frac{2\rho\kappa\nabla\alpha}{\rho_l + \rho_g} \tag{19}$$

where curvature, $\kappa$, is define as $\kappa = \nabla \cdot \frac{\nabla\alpha}{|\nabla\alpha|}$.

The governing equation of $\alpha$ can be written as:

$$\frac{\partial \alpha}{\partial t} + \frac{\partial \alpha u_i}{\partial x_j} = 0 \tag{20}$$

The turbulence model is a standard k-epsilon model with standard wall functions, and transport equations are not shown here for simplicity.

Computations are performed using the pressure-based solver in the commercial CFD code ANSYS Fluent. Belhadef et al. [31] proved that Fluent was capable of simulating this kind of two-phase flow problem. They simulated the atomization of a pressure-swirl injector, which was

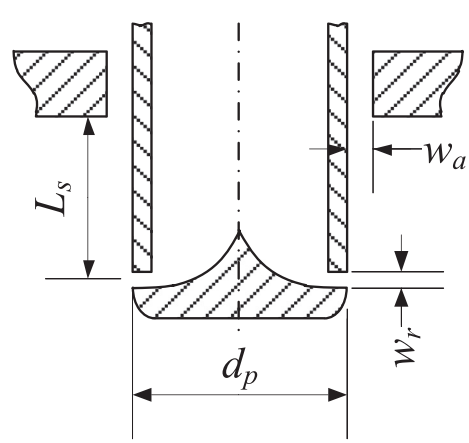


**Fig. 4.** Configuration of the present pintle injector.

**Table 1**
Operation conditions and dimensions for numerical simulations.

| $d_p$, mm | $w_r$, mm | $w_a$, mm | $L_d$, mm | $\dot{m}_{rd}$, kg/s | $\dot{m}_{ad}$, kg/s |
|---|---|---|---|---|---|
| 10 | 0.4/0.6/0.8 | 0.8/1.2/1.6 | 10 | 0.25/0.50/0.75/1.0 | 0.75/1.0/1.5 |

similar to the present problem, and the results shown good accordance with experiments.

The computational domain is a 2D axisymmetric domain, which was 5 times length of $d_p$ in radial direction and 7.3 times length of $d_p$ in axial direction, as shown in Fig. 5. Numerical simulations were conducted by using water (998kg/m$^3$, $1.003 \times 10^{-3}$ Pa s) as simulants for liquid sheet and air (1.225 kg/m$^3$, $1.789 \times 10^{-5}$ Pa s) for ambient medium. In addition, Fig. 5 shows the boundary conditions. Both axial inlet and radial inlet were set as velocity inlet conditions to match their mass flow rate listed in Table 1. Pressure outlets representing free stream conditions (1atm, 300K) were imposed at the top and right boundary faces. No-slip conditions were imposed on the wall boundaries. Computation domain was initialized full of air (1atm, 300K).

Grid independence tests were performed by using Grid-0 (80788 cells), Grid-1 (323152 cells) and Grid-2 (1292608 cells). The gas liquid interfaces ($\alpha = 0.5$) near the impinging point are shown in Fig. 6. It was found that the difference between Grid-1 and Grid-2 was almost undiscerned. Therefore, Grid-1 was chosen for the rest of computations.

### 3.2. Experiments

Cold experiments were conducted at atmospheric conditions with filtered water as simulants for both radial and axial flows. Fig. 7 shows the experimental setup, where water was pressurized by high pressure nitrogen and fed to the pintle injector. Volume flow rates were measured by turbine flow meters with 0.5% F.S. accuracy.

The backlit photography technique [32] was adopted for instantaneous spray visualization. A high speed camera (Photron Fastcam SA-X2) and a LED array (LUSTER Light Tech, 100 × 100 mm) were used. The frame rate of the camera was 5000 fps and the exposure time was 25μs.

The pintle injector for experiments is similar to that in simulations, as shown in Fig. 4. Main dimensions of the pintle injector are listed in Table 2. Effects of radial slot dimensions were considered. In total 23 experiments were carried out with $\dot{m}_{ad}$ ranging from 0.119 kg/s to 0.470 kg/s and $\dot{m}_{rd}$ ranging from 0.0337 kg/s to 0.155 kg/s.

### 3.3. Image processing

It is found that the gas-liquid interface ($\alpha = 0.5$) near the impinging point is stationary and smooth. Thus, spray angles are measured directly

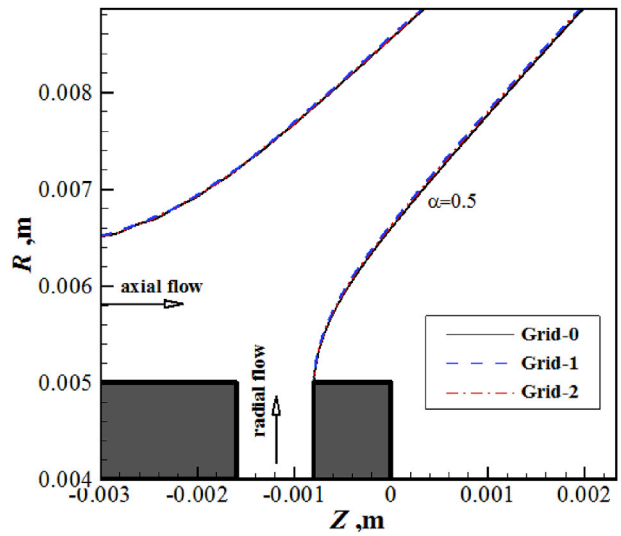


**Fig. 6.** Grid independence test.

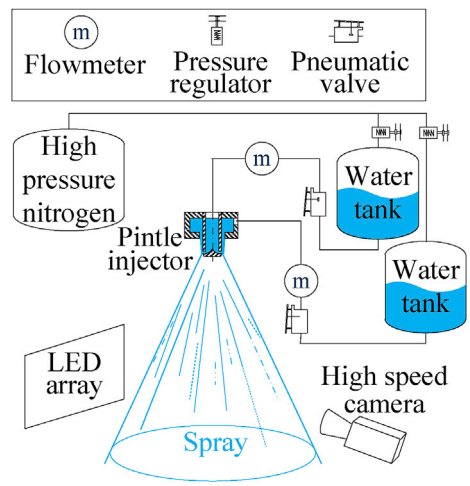


**Fig. 7.** Test rig overview with measurement devices.

**Table 2**
Main dimensions of the pintle injector in experiments.

| $d_p$, mm | $w_r$, mm | $w_a$, mm | $L_s$, mm |
|---|---|---|---|
| 14.40 | 0.48/0.60 | 1.05 | 9.5 |

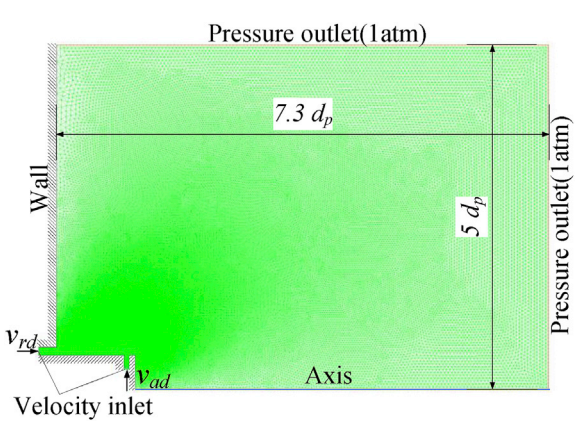


**Fig. 5.** Numerical simulation computational domain (Grid-1) and boundary conditions.

from snapshots of numerical results, as depicted in Fig. 8a. However, the spray boundary in the image obtained by high speed camera is rough and slightly varies at different moments. Therefore, the transient spray angles are obtained from experimental data by using the method proposed by Daviault et al. [33]. The image-processing procedures of each image are shown briefly in Fig. 8b. Firstly, the area ranges from the radial slot to $1.5d_p$ downstream the radial slot is selected and converted to a gray image and further converted to a binary image base on the Otsu's method [34]. In the binary image, the white pixels represent the spray whereas the black pixels represent the background. Then, the spray boundaries are picked out. At last, the boundaries are linearly fitted. And then the transient spray angle can be calculated by the slope of the lines as Equation (21).

$$2\theta = \pi - \arctan(k_r) + \arctan(k_l) \tag{21}$$

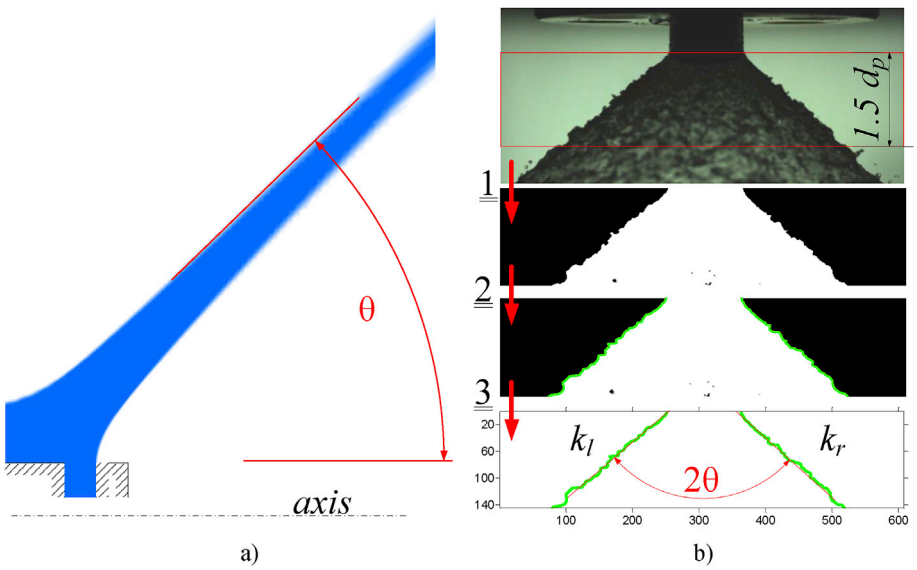


**Fig. 8.** Measuring the spray angle from (a) simulation and (b) experiment.

## 4. Model validation and discussion

According to the theoretical analysis, the spray angle is mainly dependent on the momentum ratio, as shown in Equation (16). Therefore, all spray angles obtained from both numerical and experimental results are plotted versus the momentum ratio, as shown in Fig. 9. It is noted that the spray angle is time-averaged spray angle from the spray angle time series, which collects the spray angles of the image sequence. The error bars are plotted as ±3σ, where σ is the standard deviation of the spray angle time series. It can be seen that both numerical and experimental spray angles coincide with theory well in a wide momentum ratio range. The coincidence implies that the spray angle is dominated by the momentum ratio. Thus, the theoretical model is validated. Besides, the underestimate of spray angle mentioned before is not observed, and most of experimental and numerical data are smaller than the theory predicted value. This may result from that the effects of velocity distribution at the inlet $A_{2i}$ under inviscid condition are smaller when compared with effects of viscosity of fluids.

Fig. 10 compares the spray angle obtained from different models. One model mentioned by Heister [35] indicates that the spray angle scales as $TMR^{0.5}$. The curve of this model in Fig. 10 is driven by equation $\theta = 180/\pi \cdot TMR^{0.5}$. As can be seen, differences between Heister's model and present model are less than 10° when $TMR$ is less than 1.8. This model stems from liquid jet in crossflow, and it may not be the same with the liquid-liquid sheets impingement.

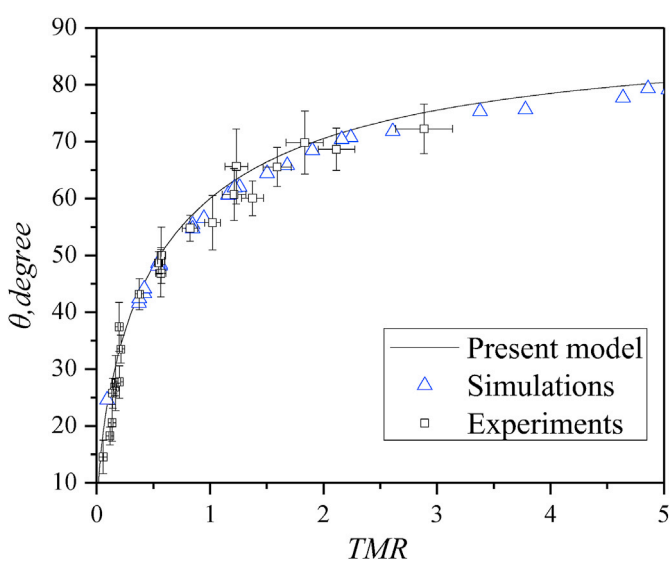


**Fig. 9.** Dependence of half spray angle on momentum ratio.

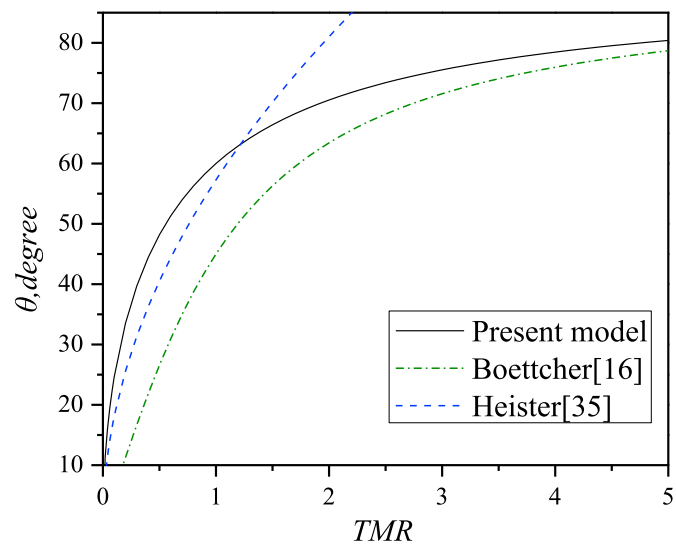


**Fig. 10.** Comparison of different models.

Boettcher et al. [16] studied a gas-gas pintle injector and proposed a model as follow:

$$\cos\theta = \frac{1}{\sqrt{1 + TMR^2}} \tag{22}$$

This equation actually satisfies global momentum conservation, which means that pintle wall has no effects on the spray angle. And it can be seen that spray angles predicted by this model are always smaller than that of present model. This difference between the liquid-liquid pintle injector and gas-gas pintle injector results from the effects of the pintle wall on the spray angle.

Fig. 11 shows pressure distributions near the pintle tip in the present study. As can be seen, a local high pressure zone locates near the radial

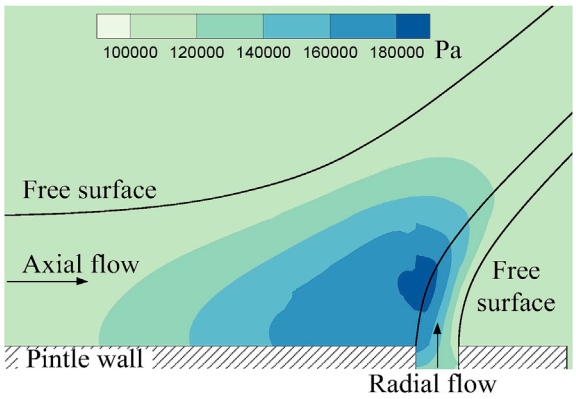


**Fig. 11.** Contours of pressure near the pintle tip.

slot. This local high pressure zone results from the impingement of the two sheets. Parts of fluid slow down and lead to an increase its static pressure. Then the force the pintle wall acts on the liquid sheet increases and becomes larger than forces act on free surface. As a result, the liquid sheets are pushed away from the pintle. Obviously, the local high pressure zone extends the spray in radial direction thus enlarges the spray angle. Unlike gas-gas pintle injectors, the pintle wall influences the spray angle of liquid-liquid pintle injectors, especially for small *TMR*, as can be seen in Fig. 10. This difference may result from the compressibility of gas. The difference between the present model and Boettcher's model decreases as *TMR* increases, which implies that the effects of the local high pressure zone becomes weaker as *TMR* increases.

## 5. Conclusions

In the present study, a spray model of pintle injectors is theoretically established to predict the spray angle. The spray model fits numerical data and experimental data reasonably well. The model indicates that the spray angle is dominated by the total momentum ratio, and other geometry parameters and operation parameters influence the spray angle through changing the momentum ratio. Numerical results show that the impingement of the two liquid sheets leads to a local high pressure zone outside the radial slot. This local high pressure zone is thought to be responsible for larger spray angles when compared with gas-gas pintle injectors. Meanwhile, the effect of local high pressure zone on spray angle becomes weaker as *TMR* increases.

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